Homework 7. Shrinkage Methods: Ridge Regression and LASSO

Stat 427/627, Statistical Machine Learning

See due date on Canvas

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16 pts. This assignment covers two Shrinkage methods: ridge regression and lasso. Although B-spline and smoothing methods are not included this assignment, be sure to review the smoothing methods in the notes and the lab.

Numerical results in this assignment may vary due to randomization.

1 College Applications (p.287, 16 pts)

Predict the number of applications received based on the other variables in the `College` data set in {`ISLR2`}.

Use the following code to split the data set randomly. Use the subset `college.data` for this assignment. Set aside `holdout.data` data for the next assignment.

```
library(ISLR2)
data("College")

my.college <- College[-484, ] # remove an extreme case.
#my.college <- College[College$Apps <=16000, ] # remove several extreme case.
train.pct <- 0.78
set.seed(2025)
Z <- sample(nrow(my.college), floor(train.pct*nrow(my.college)))
college.data <- my.college[Z, ]
holdout.data <- my.college[-Z, ]
```

Recall that we fitted the (training) data in Homework 2.

- We fit the **full** model with all 17 predictors. That is:

  ```
  college.lmF <- lm(Apps ~ ., data = college.data)
  college.lmF
  ```

- Using stepwise selection, AIC is the smallest with 11 predictors (12 parameters):

  Apps ~ Private + Accept + Top10perc + Top25perc + F.Undergrad + Outstate + Room.Board + PhD + perc.alumni + Expend + Grad.Rate,

- Using the best-subset algorithm. BIC is the smallest with 8 predictors (9 parameters):

  Apps ~ Private + Accept + Top10perc + Top25perc + Outstate + Room.Board + PhD + Expend

Continue working on the training data set (`college.data` from above) and the above models.

(a) (4 pt) Compute the Variance Inflation Factor (VIF) and comment on the severity of the collinearity of the data. Why is "collinearity" a concern, even if the model is correct?

(b) (4 pts) Evaluate prediction accuracy of your selected models based on AIC and BIC respectively. Estimate the prediction mean squared error by 10-fold cross-validation. Recall that `glm( ..., family=gaussian)` fits linear regression and its outcome can be used in `cv.glm()` (`boot` package) for cross-validation.

(c) (4 pts) Consider the **full** model with all 17 predictors (reminder: use data frame `college.data` you recreated at the beginning). Use functions in package `glmnet` to fit a ridge regression.

- Select λ chosen by (default 10-fold) cross-validation.
- Plot the results of the cross-validation.
- Report the esimated MSE of the model based on your selected λ.

(d) (4 pts) Consider the **full** model with all 17 predictors (reminder: use data frame `college.data` you recreated at the beginning). Use functions in package `glmnet` to fit a LASSO regression.

- Select λ chosen by (default 10-fold) cross-validation.
- Plot the results of the cross-validation.
- Report the esimated MSE of the model based on your selected λ.

We will continue working on this data set in the next assignment.

—— **This is the end of HW 7.** ——